

# Inflation Without an Inflaton: Observer-Screen Synchronization as an OPH Cosmology Branch

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## Abstract

OPH cosmology replaces the inflaton with finite observer-screen synchronization. Flatness, horizon coherence, near scale invariance, hot source release, and acoustic-transfer inputs are treated as jobs for the finite screen state rather than for a new scalar field.

The branch is conditional. Screen spectra, repair gaps, release amplitudes, and dark-sector inputs become physical cosmological inputs only after the source law, radial lift, transfer calculation, and likelihood comparison are fixed. Without that chain, simulator outputs are diagnostics.

## 1 Scope

This is a technical companion outside the release bundle. It states the inflation-free branch against the compact, screen-microphysics, dark-sector, and finite-source CMB papers.

## 2 Finite Quotient and Claim-Tier Firewall

**Finite quotient ensemble theorem surface.** Fix a finite regulator  $r$ . The physical presentation space is  $\Sigma_r$ , the presentation redundancy groupoid is  $\Gamma_r$ , and the finite physical quotient is

$$Q_r = \Sigma_r / \Gamma_r, \quad \pi_r : \Sigma_r \rightarrow Q_r.$$

The quotient removes nonphysical presentation data: gauge representatives, port relabelings, mesh labels, shard or worker identifiers, queue order, repair schedule identifiers, retry counters, timestamps unless declared semantic, hidden carrier coordinates, and inert ancillary labels. If only settled configurations carry probability, the probability space is the normal-form subset

$$N_r = n_r(Q_r).$$

The map  $n_r$  is a normal-form map, not a probability law. Any promoted physical branch inherits this firewall: it must declare the quotient-intrinsic source law or action before a normal form can be read as a selection or prediction claim.

**Observable algebras and reference states.** In the finite classical case the quotient observable algebra is

$$\mathcal{O}_r = \ell^\infty(Q_r).$$

In the finite quantum case the physical algebra is a declared quotient algebra  $\mathcal{A}_r^{\text{phys}}$  with state

$$\omega_r(A) = \text{Tr}(\rho_r A).$$

When the reference object is obtained from a finite lifted carrier, the load-bearing data are not an abstract groupoid cardinality alone. They are a tracially pointed quotient

$$(\mathcal{A}_{r,b}^{\text{phys}}, \tau_{r,b}^0), \quad \mathcal{A}_{r,b}^{\text{phys}} = z_{r,b} B(\tilde{\mathcal{H}}_r)^{G_r} z_{r,b},$$

where  $U_r : G_r \rightarrow U(\tilde{\mathcal{H}}_r)$  is the compact gauge action and  $z_{r,b}$  is the central projection for the declared boundary or superselection sector. The reference trace is

$$\tau_{r,b}^0(A) = \frac{\text{Tr}_{\tilde{\mathcal{H}}_r}(A)}{\text{Tr}_{\tilde{\mathcal{H}}_r}(z_{r,b})}.$$

If

$$z_{r,b} \tilde{\mathcal{H}}_r \cong \bigoplus_{\alpha} V_{\alpha} \otimes M_{\alpha},$$

with  $d_{\alpha} = \dim V_{\alpha}$  and  $m_{\alpha} = \dim M_{\alpha}$ , then

$$\mathcal{A}_{r,b}^{\text{phys}} \cong \bigoplus_{\alpha} I_{V_{\alpha}} \otimes B(M_{\alpha}), \quad p_{r,\alpha} = \frac{d_{\alpha} m_{\alpha}}{\sum_{\beta} d_{\beta} m_{\beta}}.$$

These are the induced central-sector weights only after the carrier representation and boundary sector have been fixed.

**OPH quotient ensemble.** An OPH quotient ensemble is specified by a quotient-intrinsic base weight and action

$$m_r : Q_r \rightarrow \mathbb{R}_{>0}, \quad S_r : Q_r \rightarrow \mathbb{R} \cup \{+\infty\},$$

and

$$w_r(q) = m_r(q) e^{-S_r(q)}, \quad Z_r = \sum_{q \in Q_r} w_r(q), \quad \mu_r(q) = Z_r^{-1} w_r(q).$$

Equivalently, one may state an intrinsic projective prior  $\nu_r$  on  $Q_r$  and set  $\mu_r = (n_r)_{\#} \nu_r$ . Uniform quotient counting, uniform representative counting pushed to the quotient, groupoid weights, and tracial central-sector weights are different physical claims. The paper must declare which one is being used.

**Normal-form projector non-selection.** For any map  $N : Q \rightarrow Q_{\text{nf}}$ , the induced map on laws

$$\mathcal{C}_Q(\mu) = N_{\#} \mu$$

is idempotent:

$$\mathcal{C}_Q^2 = \mathcal{C}_Q.$$

Every law supported on  $Q_{\text{nf}}$  is fixed. Therefore settlement or canonicalization never selects a unique physical probability law by itself.

**Selection-gap corollary.** Let  $X \subseteq Q_{\text{nf}}$  be a finite set of quotient-normal candidates distinguished by visible invariants. Normal-form data determine  $X$  and its quotient-visible invariants, but they do not choose a member of  $X$ . If two laws  $\mu, \nu$  are supported on  $X$  and concentrate on different candidates, both are fixed by  $\mathcal{C}_Q$ . Unique sector selection therefore requires source data: an intrinsic action with a unique minimizer, a declared physical ensemble, or a refinement-stable gap certificate. A defect or holonomy classification can classify possible sectors without choosing the physical sector, and a contraction or repair generator can certify convergence toward a declared target without creating the target law.

**Finite MaxEnt quotient ensemble.** For finite  $Q$ , positive  $m$ , and quotient observables  $F_1, \dots, F_k$ , maximizing

$$\mathcal{H}_m(\nu) = - \sum_q \nu(q) \log \frac{\nu(q)}{m(q)}$$

subject to

$$\sum_q \nu(q) = 1, \quad \sum_q \nu(q) F_a(q) = c_a$$

has the full-support solution, when the feasible full-support surface is nonempty,

$$\mu(q) = \frac{m(q) \exp[-\sum_a \theta_a F_a(q)]}{Z(\theta)}.$$

Boundary optima obey the same formula after restricting to their support. On a finite noncommutative quotient algebra with faithful reference state  $\sigma_r$ ,

$$\rho_r = \frac{\exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}{\text{Tr} \exp(\log \sigma_r - \sum_a \theta_a F_{r,a})}.$$

The finite constraint ledger must name every  $F_{r,a}$ , its units and support, the target expectation and source, sector or zero-mode treatment, refinement transformation, and proof that no run output or observational output entered the source definition.

**Refinement compatibility and RG closure.** For  $s \succeq r$ , let  $c_{sr} : Q_s \rightarrow Q_r$  be the physical coarse map. Exact compatibility of weighted ensembles is equivalent to the fiber-sum identity

$$\sum_{q': c_{sr}(q')=q} m_s(q') e^{-S_s(q')} = \alpha_{sr} m_r(q) e^{-S_r(q)}$$

for a constant  $\alpha_{sr} > 0$  independent of  $q$ . Then

$$(c_{sr})_{\#} \mu_s = \mu_r.$$

If the one-step defects are

$$\delta_{k+1,k} = \|(c_{k+1,k})_{\#} \mu_{k+1} - \mu_k\|_{\text{TV}},$$

then

$$\|(c_{nr})_{\#} \mu_n - \mu_r\|_{\text{TV}} \leq \sum_{k=r}^{n-1} \delta_{k+1,k}.$$

For exponential-family refinement, exact closure requires the fine conditional free energy

$$G_{sr,\theta}(q_r) = -\log \mathbb{E}_{m_s^0} [\exp[-\theta \cdot F_s(Q_s)] \mid c_{sr}(Q_s) = q_r]$$

to equal  $\kappa_{sr}(\theta) + R_{sr}(\theta) \cdot F_r(q_r)$ . If the residual is uniformly bounded by  $\varepsilon$ , the induced total-variation defect is bounded by  $\tanh \varepsilon$ .

**Implementation invariance and representative lifting.** If implementations  $A, B$  have quotient bijections  $h_r : Q_r^A \rightarrow Q_r^B$  satisfying

$$m_r^B(h_r q) = m_r^A(q), \quad S_r^B(h_r q) = S_r^A(q), \quad h_r \circ c_{sr}^A = c_{sr}^B \circ h_s,$$

then

$$(h_r)_\# \mu_r^A = \mu_r^B.$$

For tracially pointed quantum quotients the corresponding equivalence is a trace-preserving quotient equivalence. It is invariant under unitary intertwiners preserving the gauge action and sector, and under inert trivial ancillas  $A \mapsto A \otimes I_{\text{anc}}$ . It is not invariant under arbitrary changes of gauge-representation multiplicities.

If an implementation stores representatives, a representative-level law must be a conditional lift

$$\tilde{\mu}_r(x) = \mu_r(\pi_r x) \kappa_r(x \mid \pi_r x), \quad \sum_{x: \pi_r(x)=q} \kappa_r(x \mid q) = 1.$$

Then  $(\pi_r)_\# \tilde{\mu}_r = \mu_r$ . Uniform representative sampling yields orbit-size weights and is physical only if representative counting is the declared base measure.

**Quotient-lumpable kernels and sampler correctness.** A representative kernel  $\tilde{P}(x, y)$  descends to  $Q_r$  only when

$$P_Q(q, q') = \sum_{y: \pi(y)=q'} \tilde{P}(x, y)$$

is independent of the chosen representative  $x \in \pi^{-1}(q)$ . For  $w(q) = m(q)e^{-S(q)}$  and proposal  $R(q, q')$  with reciprocal support, the Metropolis–Hastings acceptance rule

$$a(q, q') = \min \left\{ 1, \frac{w(q')R(q', q)}{w(q)R(q, q')} \right\}$$

gives detailed balance

$$\mu(q)R(q, q')a(q, q') = \mu(q')R(q', q)a(q', q).$$

Repair-informed proposals must include the Hastings asymmetry term; otherwise the stationary law is generically changed.

**Repair generators are not selectors.** A repair generator of the form

$$L_{\text{rep}} = \sum_C c_C (I - E_C)$$

is a relaxation or sampling object after a law has been selected. Conditional expectations  $E_C$  are defined on  $L^2(X_r, \pi_r)$ , so the reference law  $\pi_r$  is input. On overlapping collars the expectations need not commute. The correct finite gap certificate is the Poincare constant

$$\kappa_r = \inf_{f \perp 1} \frac{\sum_C \|(I - E_C)f\|^2}{\|f\|^2}.$$

If local fiber rates have a positive lower bound  $\gamma_*$ , then

$$L_{\text{rep}} \geq \gamma_* \kappa_r (I - P_0).$$

Finite repair completeness gives  $\kappa_r > 0$  at fixed regulator. A uniform refinement lower bound  $\inf_r \kappa_r > 0$  is a separate theorem or receipt.

**Finite evidence accuracy.** For bounded coarse observables  $O$ , if

$$\|\hat{\mu}_s - \mu_s\|_{\text{TV}} \leq \epsilon_{\text{samp}}$$

and the refinement defects sum to  $\epsilon_{\text{ref}}$ , then

$$\left| \mathbb{E}_{\hat{\mu}_s}[O \circ c_{sr}] - \mathbb{E}_{\mu_r}[O] \right| \leq 2\|O\|_{\infty}(\epsilon_{\text{samp}} + \epsilon_{\text{ref}}).$$

Continuum-facing observables require a realization map and correlation Cauchy bound in addition to a finite histogram.

**Vacuum promotion gate.** A stationary sampler is not a physical vacuum. For any faithful target law one can build a positive transfer operator with that law as ground state, so positivity alone is not a selector. Vacuum promotion requires source Euclidean slab data

$$\mathfrak{S}_r^E = (Q_r, m_r^0, J_r, V_r, a_{t,r})$$

whose conductance  $J_r(q, q') = J_r(q', q) \geq 0$ , local potential  $V_r$ , and slab thickness  $a_{t,r}$  are derived without using the target law or sampler output. With connected event graph,

$$(H_r^E f)(q) = \frac{1}{m_r^0(q)} \sum_{q'} J_r(q, q') (f(q) - f(q')) + V_r(q) f(q)$$

is self-adjoint and bounded below on  $L^2(Q_r, m_r^0)$ ; its finite Feynman–Kac semigroup is positivity improving. Perron–Frobenius gives a unique positive normalized ground state  $\Omega_r$ , and the finite vacuum law is

$$\mu_r^{\text{vac}}(q) = |\Omega_r(q)|^2 m_r^0(q).$$

For  $T_r = e^{-a_{t,r}(H_r^E - E_{0,r})}$ , the Doob kernel is stochastic and detailed-balanced with  $\mu_r^{\text{vac}}$ . Continuum promotion additionally requires reflection positivity or equivalent reconstruction plus refinement compatibility of the transfer family.

**Primordial and cosmological prediction firewall.** A screen covariance is not automatically a primordial curvature spectrum. The map

$$C_\ell = 4\pi \int_0^\infty \frac{dk}{k} \Delta_\zeta^2(k) j_\ell^2(k\chi_\star)$$

has a null space unless a radial prior, finite parametric family, or source-stress bridge is declared. OPH primordial promotion requires the source-only stress, single-clock, entropy-repair, curvature-evolution, adiabatic-mode, phase-coherence, screen-to-radial-lift, radial-null-space, and forward-projection receipts. Observable CMB promotion further requires frozen source, solver, likelihood, no-data-use, pooled-reducer, and falsification-rule receipts.

**Claim tiers and required receipts.** Every ensemble-facing run records immutable ensemble id, claim tier, regulator id, representative schema hash, gauge action hash, canonicalizer hash, base measure, action coefficients, coarse maps, zero-mode projector, amplitude convention, sampler hash,

smoothing policy, source provenance, and explicit nonclaims. The seed belongs to the run receipt, not to the ensemble definition. The claim tiers are

- $E0$  : seed noise, proposal noise, repair jitter,
- $E1$  : conventional reference ensemble,
- $E2$  : OPH-native quotient ensemble,
- $E3$  : OPH vacuum,
- $E4$  : OPH primordial field,
- $E5$  : observable cosmological prediction.

The evidence bundle must keep separate receipts for stationary-law schedule invariance, detailed balance of the aggregate kernel, and pathwise partition invariance. Deterministic replay of semantic random streams or a canonical serial chain is useful, but it is not pathwise partition invariance. Smoothing must preserve raw coefficients, raw spectra, smoothing kernels, smoothed coefficients, smoothed spectra, and hashes of each stage; it is not part of  $S_r$  unless explicitly declared.

### 3 Finite Screen Spectrum Theorem Package

This fragment owns the screen-level spectrum theorem used by the staged cosmology branch. It does not own TT, TE, EE, lensing, likelihoods, or physical CMB promotion. Those remain downstream Boltzmann and data-contract gates.

**Definition 3.1** (finite screen regulator). *For regulator  $r$ , a finite screen regulator is*

$$\mathfrak{S}_r = (\mathcal{T}_r, M_r, \Gamma_r, \mathcal{Q}_r, C_{sr}, J_{rs}),$$

where  $\mathcal{T}_r$  is a finite cellulation of the screen,  $M_r \succ 0$  is the area/quadrature mass matrix,  $\Gamma_r$  is the hidden-presentation groupoid,  $\mathcal{Q}_r = \Sigma_r/\Gamma_r$  is the physical quotient, and  $C_{sr}$ ,  $J_{rs}$  are the coarse-graining and interpolation maps. Shape regularity and quadrature convergence require

$$f^\top M_r g \rightarrow \int_{S^2} f g d\Omega$$

with an explicit band-limited error bound

$$\left| f^\top M_r g - \int f g d\Omega \right| \leq \varepsilon_{M,r}(L) \|f\|_{H^s} \|g\|_{H^s}.$$

Patch count alone is therefore not an angular-resolution certificate.

**Definition 3.2** (geometric screen scalar). *Let the quotient-visible collar-volume readout be*

$$J_{X,r}(x) = \lambda_r(x) \sqrt{\det \sigma_{AB,r}(x)} > 0,$$

and let  $\bar{J}_{X,r}$  be emitted by an independently defined homogeneous or frozen-background branch, not by a CMB fit. Define

$$q_{0,r} = \frac{1}{3} \log \frac{J_{X,r}}{\bar{J}_{X,r}}.$$

On a certified spherical chart, with  $B_r = [1, n_x, n_y, n_z]$ ,

$$\Pi_{<2,r} = B_r (B_r^\top M_r B_r)^{-1} B_r^\top M_r, \quad q_r = (I - \Pi_{<2,r}) q_{0,r}.$$

On an irregular screen,  $B_r$  is replaced by the certified generalized eigenprojector onto the finite  $\ell = 0, 1$  subspace. Hard-coded feature  $z$ -scores or observer-summary weights are diagnostic proxies, not  $q_r$ .

**Proposition 3.3** (quotient invariance). *If a recharting  $U$  preserves the screen inner product and geometric readout,*

$$U^\top M'_r U = M_r, \quad q'_{0,r} = U q_{0,r}, \quad B'_r = U B_r,$$

*then  $\Pi'_{\geq 2,r} U = U \Pi_{\geq 2,r}$ , and  $q'_r = U q_r$ .*

*Proof.* Substitution gives

$$\begin{aligned} \Pi'_{< 2,r} U &= U B_r (B_r^\top U^\top M'_r U B_r)^{-1} B_r^\top U^\top M'_r U \\ &= U B_r (B_r^\top M_r B_r)^{-1} B_r^\top M_r = U \Pi_{< 2,r}. \end{aligned}$$

Subtracting from  $U$  gives the high-mode identity.  $\square$

**Definition 3.4** (physical scalar precision and action). *The scalar precision  $K_r$  must have a physical origin on  $V_r = \text{im } \Pi_{\geq 2,r}$ . Allowed branches are:*

$$K_r^{\text{Hess}} = \nabla^2 \Phi_r \Big|_{q=0},$$

*for a quotient-visible mismatch or release free energy; a repair Dirichlet form*

$$K_r^{\text{diss}} = H_r - T_r^\top H_r T_r, \quad T_r^\top H_r T_r \preceq H_r;$$

*or the reversible generator form*

$$K_r^{\text{Dir}} = -\frac{1}{2}(L_r + L_r^{\dagger H}).$$

*A raw nonsymmetric repair matrix or caller-supplied  $\kappa$  is not a precision operator. The absolute normalization must be fixed independently, for example by*

$$K_0 Y_{\ell m} = \ell(\ell+1) Y_{\ell m}$$

*on the  $\theta = 0$  branch. Once normalized,*

$$S_{\text{scr},r}[q] = \frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r = \frac{1}{2A_{q,r}} q^\top M_r K_r q.$$

**Theorem 3.5** (finite MaxEnt screen covariance). *Let  $K_r e_i = \kappa_i e_i$  with an  $M_r$ -orthonormal basis of  $V_r$ ,  $\kappa_i > 0$ , and  $d_r = \dim V_r$ . Among continuous densities in coefficients  $q = \sum_i a_i e_i$  with  $\mathbb{E}[a_i] = 0$  and*

$$\frac{1}{2} \mathbb{E} \langle q, K_r q \rangle_r = E_{q,r}^{\text{src}},$$

*the unique entropy maximizer is*

$$d\mu_r(q) = Z_r^{-1} \exp \left[ -\frac{1}{2A_{q,r}} \langle q, K_r q \rangle_r \right] dq, \quad A_{q,r} = \frac{2E_{q,r}^{\text{src}}}{d_r},$$

*and*

$$\mathbb{E}[a_i a_j] = \delta_{ij} \frac{A_{q,r}}{\kappa_i}.$$

*Proof.* The Euler–Lagrange equation for entropy with normalization and quadratic-energy constraints gives a Gaussian density with inverse temperature  $\beta_r$ . The expected energy is

$$E_{q,r}^{\text{src}} = \frac{1}{2} \sum_i \kappa_i \frac{1}{\beta_r \kappa_i} = \frac{d_r}{2\beta_r}.$$

Thus  $A_{q,r} = \beta_r^{-1} = 2E_{q,r}^{\text{src}}/d_r$ . Strict concavity of entropy gives uniqueness.  $\square$

*Remark 3.6* (discrete quotient branch). For a discrete finite quotient with intrinsic base weight  $m_r(q)$ , the exact Gibbs law is

$$\mu_r(q) = Z_r^{-1} m_r(q) e^{-\beta_r E_r(q)}.$$

A Gaussian statement on that branch requires a separate Laplace or central-limit theorem with controlled higher-order terms.

**Definition 3.7** (source release energy). *The amplitude source is an independently selected quotient ensemble  $\nu_r^{\text{src}}$  on frozen or released collar normal forms:*

$$E_{q,r}^{\text{src}} = \frac{1}{2} \int \langle q_r(x), K_r q_r(x) \rangle_r d\nu_r^{\text{src}}(x).$$

*The receipt must expose  $\nu_r^{\text{src}}$ , the base measure,  $K_r$ ,  $d_r$ ,  $E_{q,r}^{\text{src}}$ ,  $A_{q,r}$ , no-observation ancestry, and the same operator normalization used in the action. MaxEnt alone does not determine amplitude.*

**Theorem 3.8** (rotational screen spectrum). *If the continuum precision  $K_\theta$  commutes with the  $SO(3)$  action on scalar functions, then by Schur's lemma each  $\mathcal{H}_\ell$  is an eigenspace. For the exact conformal-shell family,*

$$\Lambda_\ell(\theta) = \frac{\Gamma(\ell + 2 + \theta/2)}{\Gamma(\ell - \theta/2)}, \quad \ell \geq 2.$$

*Then  $\Lambda_\ell(0) = \ell(\ell + 1)$ , and the MaxEnt covariance gives*

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

*Consequently  $\mathcal{D}_\ell^q = \ell(\ell + 1)C_\ell^q/(2\pi) \sim (A_q/2\pi)\ell^{-\theta}$ .*

**Theorem 3.9** (finite refinement error). *Suppose that for  $2 \leq \ell \leq L$*

$$\left| \frac{\lambda_{\ell m,r}}{\Lambda_\ell(\theta)} - 1 \right| \leq \varepsilon_{K,r}(L), \quad \left| \frac{A_{q,r}}{A_q} - 1 \right| \leq \varepsilon_{A,r}, \quad \varepsilon_{K,r} < 1.$$

*Then  $C_{\ell m,r}^q = A_{q,r}/\lambda_{\ell m,r}$  satisfies*

$$\left| \frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell(\theta)} - 1 \right| \leq \frac{\varepsilon_{A,r} + \varepsilon_{K,r}}{1 - \varepsilon_{K,r}}.$$

*Proof.* Write  $A_{q,r} = A_q(1 + a_r)$  and  $\lambda_{\ell m,r} = \Lambda_\ell(1 + b_{\ell m,r})$ . Then

$$\frac{C_{\ell m,r}^q}{A_q/\Lambda_\ell} = \frac{1 + a_r}{1 + b_{\ell m,r}},$$

and the displayed bound follows from  $|a_r| \leq \varepsilon_A$ ,  $|b_{\ell m,r}| \leq \varepsilon_K$ . □

**Theorem 3.10** (refinement-semigroup tilt). *Let  $R_b$  be the scalar refinement map for scale ratio  $b > 1$ . If one isolated scalar covariance mode has positive eigenvalue  $\lambda(b)$ ,  $\lambda(b_1 b_2) = \lambda(b_1)\lambda(b_2)$ , and  $\lambda$  is continuous, then there is a unique real  $\theta$  with*

$$\lambda(b) = b^{-\theta}, \quad \theta = -\frac{\log \lambda(b)}{\log b}.$$



*Proof.* Set  $g(t) = -\log \lambda(e^t)$ . The semigroup law gives  $g(t+s) = g(t) + g(s)$ . Continuity makes  $g(t) = \theta t$ , hence the result.  $\square$

**Target 3.11** (edge-center  $P_\star/48$  tilt). *The identity  $\theta = P_\star/48$  is a theorem-grade statement only after three lemmas pass: an oriented edge-center reserve  $\rho_{\text{full}} = P_\star/24$ , a scalar half-collar projector  $\rho_q = P_\star/48$ , and a reserve-to-RG Ward identity equating this reserve with  $-d \log \mathcal{D}_\ell^q / d \log \ell$ . Until then  $P_\star/48$  and  $e(P_\star - \varphi)$  are distinct diagnostic hypotheses.*

**Theorem 3.12** (thin-shell power-law lift). *Assume  $q(\hat{n}) = Z_\star \Pi_{\ell \geq 2} \zeta_\star(R_\star \hat{n})$  and*

$$\Delta_\zeta^2(k) = A_\zeta (k/k_\star)^{-\theta}.$$

*Then*

$$C_\ell^q = 4\pi Z_\star^2 \int d \ln k \Delta_\zeta^2(k) j_\ell^2(k R_\star),$$

*and, for  $-2 < \theta < 4$ ,*

$$A_q = \pi^{3/2} Z_\star^2 A_\zeta (k_\star R_\star)^\theta \frac{\Gamma(1 + \theta/2)}{\Gamma(3/2 + \theta/2)}.$$

*Thus*

$$A_\zeta = \frac{A_q}{\pi^{3/2} Z_\star^2 (k_\star R_\star)^\theta} \frac{\Gamma(3/2 + \theta/2)}{\Gamma(1 + \theta/2)}.$$

**Proposition 3.13** (finite-window bound). *Let  $\Psi_\ell(k) = \int dr W(r) j_\ell(kr)$ ,  $W \geq 0$ ,  $\int W dr = 1$ , with mean  $R_\star$  and variance  $\sigma_R^2$ . If*

$$\delta_\ell(k) = \frac{k^2 \sigma_R^2}{2} \sup_{r \in \text{supp } W} |j_\ell''(kr)|,$$

*then*

$$|C_{\ell,W}^q - C_{\ell,\text{shell}}^q| \leq 4\pi Z_\star^2 \int d \ln k \Delta_\zeta^2(k) \left[ 2\delta_\ell(k) + \delta_\ell(k)^2 \right].$$

**Proposition 3.14** (radial non-identifiability). *For a finite radial basis,  $C = Ap$ . If  $A \in \mathbb{R}^{N_\ell \times N_k}$  has rank  $r$ , then  $\dim \ker A = N_k - r$ , and every  $p+v$  with  $v \in \ker A$  gives the same screen spectrum. Radial promotion therefore requires either a source theorem restricting  $p$ , or a declared prior with a published null basis, resolution kernels, positivity checks, and prior-sensitivity report.*

**Target 3.15** (screen-spectrum receipt set). *The theorem-grade screen spectrum requires the following receipt families before any primordial or CMB promotion:*

1. *geometry, scalar quotient, low-mode projector, scalar precision, and operator normalization;*
2. *quotient ensemble selection, scalar release energy, and MaxEnt screen covariance;*
3. *scalar refinement tilt and angular screen spectrum with a finite error budget;*
4. *radial window, radial null-space, forward residual, and transfer firewall.*

## 4 Branch Thesis

The branch claim is:

On a finite-source branch, OPH may replace the logical jobs of inflation by direct/CMH spatial-holonomy selection, same-boundary or low- $k$  repair coherence, a geometric screen spectrum, MaxEnt release, and ordinary Boltzmann transfer.

This claim is not part of the recovered SM/GR core. It becomes predictive only when the source objects are fixed before observational comparison.

Coherent-matter same-channel forcing supplies a local scalar-channel identity for nondegenerate OPH-coherent material sources under Scalar Edge-Center Exhaustion. It does not supply the finite covariant dark-stress parent needed for cosmological  $w_A$ ,  $c_{s,A}^2$ ,  $\sigma_A$ ,  $Q_A^\mu$ , recipient-stress, gauge-variable, CDM-limit, and refinement-convergence receipts.

The anomaly abundance selector is owned by `cosmology/oph_dark_matter_paper.tex`. The finite-source branch may use a source-only  $\bar{\rho}_A(a)$  only when

ANOMALY\_ABUNDANCE\_SOURCE\_RECEIPT

passes. Otherwise the dark continuation is either `OFF` or `CONDITIONAL_SOURCE_STATE`.

## 5 Paper Targets

| Target                | Required theorem or receipt                                                                                                                                                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Finite early state    | Quotient normal-form branch with declared base measure and implementation invariance.                                                                                                                                                                         |
| Flatness              | Zero clock-slice spatial Levi–Civita holonomy identifies $\kappa = 0$ . Exact selection requires <code>DIRECTTHEOREM</code> , <code>CONDITIONAL-CMH</code> , or <code>EXPLICITASSUMPTION</code> ; damping is only an approximate $ K $ or $ \Omega_K $ bound. |
| Horizon coherence     | Same-boundary unique-extension theorem or a uniform low- $k$ repair gap.                                                                                                                                                                                      |
| Near scale invariance | Geometric screen scalar, normalized scalar precision, source release energy, refinement tilt, and angular screen-spectrum receipts.                                                                                                                           |
| Amplitude             | Source release-energy certificate for $A_q$ , followed by a separate radial lift for $A_\zeta$ ; neither may be fitted from CMB data.                                                                                                                         |
| Hot start             | MaxEnt release into the realized radiation branch with a common release clock.                                                                                                                                                                                |
| Acoustic transfer     | Frozen source functions and Boltzmann transfer owned by the finite-source CMB and data-contract papers.                                                                                                                                                       |

## 6 Simulator Contract

The simulator reference is <https://github.com/muellerberndt/oph-physics-sim>. For this paper the simulator may emit diagnostic screen spectra, observer-screen synchronization receipts,

and release-amplitude candidates. It must not label them inflation replacement, OPH primordial field, or physical CMB prediction unless the E4 and E5 receipts in the shared theorem fragment pass in the concrete run bundle.

## 7 Flatness Boundary

The flatness branch is split into identification and selection. On a clocked FLRW continuation with  $u_a = -N\nabla_a\chi$ ,  $h_{ab} = g_{ab} + u_a u_b$ , and  $K(\tau) = \kappa/a(\tau)^2$ , the area-normalized small-loop holonomy of the spatial Levi-Civita connection identifies  $K = 0$  and hence  $\kappa = 0$ . It does not select that sector from the raw FLRW fiber. The simulator must therefore report one of `DIRECTTHEOREM`, `CONDITIONALCMH`, `EXPLICITASSUMPTION`, or `OPENTHEOREM`. S3 screen/collar defect decay is a repair diagnostic, not a spatial curvature holonomy. Curvature damping can certify only an `APPROXIMATEBOUND` on  $|K|$  or  $|\Omega_K|$ .

## 8 Screen-Spectrum Branch

The load-bearing screen result is the angular screen covariance, not a CMB spectrum:

$$C_\ell^q = A_q \frac{\Gamma(\ell - \theta/2)}{\Gamma(\ell + 2 + \theta/2)}.$$

This statement requires the geometric  $q_r$  of Definition 3.2, a normalized positive precision operator  $K_r$ , a source-only release ensemble and  $E_{q,r}^{\text{src}}$ , the MaxEnt covariance theorem, and a finite-to-continuum refinement error budget. The value  $\theta = P_\star/48$  is theorem-grade only if the edge-center reserve, half-collar scalar projection, and reserve-to-RG Ward identity in Target 3.11 pass. Otherwise  $P_\star/48$  and  $e(P_\star - \varphi)$  are distinct diagnostic hypotheses.

The primordial power law

$$\Delta_\zeta^2(k) = A_\zeta \left( \frac{k}{k_\star} \right)^{-\theta}$$

enters this paper only after the thin-shell or finite-window lift fixes  $R_\star$ ,  $Z_\star$ , the radial window, the null-space report, the forward-projection residual, and the amplitude conversion from  $A_q$  to  $A_\zeta$ . A screen theorem by itself does not prove TT, TE, EE, lensing, acoustic peaks, or an official likelihood result.

## 9 Unclosed Claim Boundaries

1. Prove the same-boundary or low- $k$  coherence branch in quotient language.
2. Replace diagnostic feature-weight scalar readouts with the geometric collar-volume  $q_r = \Pi_{\geq 2,r}(1/3) \log(J_{X,r}/\bar{J}_{X,r})$ .
3. Specify the source-release-energy certificate for  $A_q$  without CMB data use.
4. Prove or bound the refinement source of  $\theta$ , including the separate  $P_\star/48$  edge-center reserve lemmas if that branch is retained.
5. Connect the synchronized release state to the source-only primordial bridge.
6. Hand the resulting frozen inputs to the CMB and likelihood-contract papers.